ASSIGNMENT 1

Communications Systems

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Exercises

- Error Detection
- CRC
- Error Correction: Decision regions
- Error Correction: performance
- Error Correction: Syndrome decoding
- Properties of m-sequences

Exercise 1

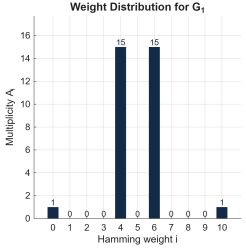
Error Detection – Two Linear Codes

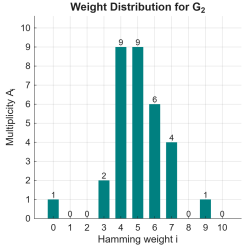
Exercise 1 - Part 1

Codebooks and Weight Spectra

4. Compute the multiplicities $A_i = \#\{c \in C : w_H(c) = i\}, \quad i = 0, \ldots, 10.$

- The multiplicities indicate the number of codewords with a given Hamming weight, since we have $n = k + r = 10$ bits we have 11 instances of multiplicities from 0 to 10
- Two generator matrices were used so there are two sets of 2^k codewords each with its own sets of multiplicities



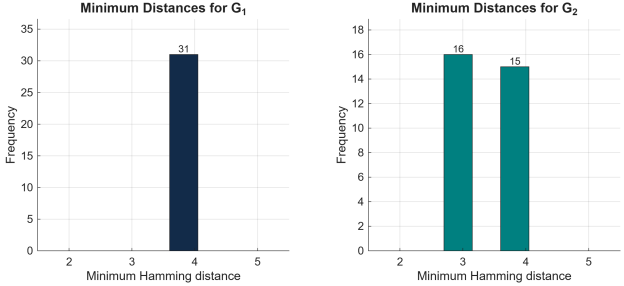


Code 1 (G_1)	
i	A_i
0	1
1	0
2	0
3	0
4	15
5	0
6	15
7	0
8	0
9	0
10	1

Code 2 (G_2)	
i	A_i
0	1
1	0
2	0
3	2
4	9
5	9
6	6
7	4
8	0
9	1
10	0

5. Determine the minimum d_{min} Hamming distance and its multiplicity A_{min}.

- The minimum Hamming distance is the lowest distance possible from one codeword compared to all the others excluding the codewords at distance 0
- Since there are 2^k codewords the maximum multiplicity theoretically would be $32 - 1 = 31$ since the last vector is not computed due to the fact that the comparison would be redundant
- The first set of codewords reaches it since all the codewords have the same min distance
- While the second set has two different min distances with different multiplicities

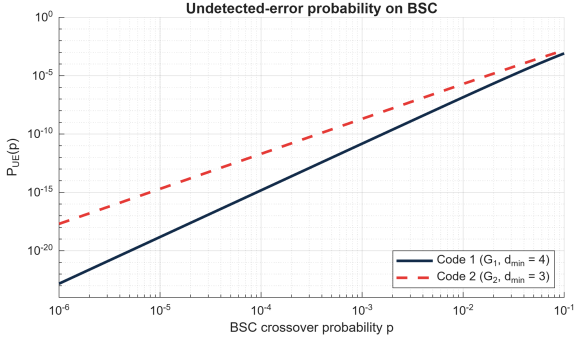


Exercise 1 - Part 2

Analytical Performance

2. Plot both curves on the same log-log axes for $p \in [10^{-6}, 10^{-1}]$

- The plot shows 2 curves (with different generator matrix) of the probability of undetected error (PUE) ranging from a probability of 10^{-6} to 10^{-1}



3. For small p, find the leading term of the two P_{UE} expressions and explain its relationship with $d_{\min}$ and $A_{\min}$.

- For small probabilities, the '$(1-p)$' part of the sum can be discarded and the new formula contains only the multiplicity part multiplied by probability.
- For linear codes, the $d_{\min}$ between c_1 and c_2 is equivalent to the result of the XOR between the two codewords that is the Hamming weight.

4. Compare the two codes: which one has better error-detection performance, and why?

- Code 1 is better because its curve lies below Code 2's curve, meaning that for a fixed bit error probability p, it achieves a lower P_{UE}.
- As you can see in the figure, Code 1 rather than Code 2 has a less rapid (steeper) curve that means better performance; for a logarithmic graph the $d_{\min}$ leads the gradient because $\log(A_{\min})$ is a constant.
- $\log(P_{UE}(p)) = \log(A_{\min}) + d_{\min} \cdot \log(p)$
- For small p ($p \to 0$) the formula can be approximated to $P_{UE} \approx A_{d_{\min}} \cdot p^{d_{\min}}$, so:
  - For G_1 the first non-null term is for $i = 4 \implies A_{\min} = 15 \implies$ the main term $P_{UE} \approx 15p^4$
  - For G_2 the first non-null term is for $i = 3 \implies A_{\min} = 2 \implies$ the main term $P_{UE} \approx 2p^3$

Exercise 1 - Part 3

Syndrome Detector

1. Construct the parity check matrix H^T using the convention of the lectures:

$$H^T = \begin{bmatrix} P \\ I_5 \end{bmatrix}, \quad H^T \in \mathbb{F}_2^{10\times 5}.$$

$$H^T = \left[\begin{array}{ccccc} 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 & 1 \\ \hline 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{array}\right] \begin{array}{l} \left.\vphantom{\begin{array}{c}1\\1\\1\\1\\1\end{array}}\right\} P\ (5\times 5) \\ \left.\vphantom{\begin{array}{c}1\\1\\1\\1\\1\end{array}}\right\} I_5\ (5\times 5) \end{array}$$

2. Provide two received vectors y_1 and y_2 with zero and non-zero syndromes, respectively. Explain your choices.

- **Zero syndrome:** Encode message $m = 1\ 0\ 0\ 1\ 0$ directly via $y_1 = m \cdot G_1 \bmod 2$:

$$y_1 = 1\ 0\ 0\ 1\ 0 \mid 0\ 1\ 1\ 1\ 1$$

$$s_1 = y_1 \cdot H^T \bmod 2 = 0\ 0\ 0\ 0\ 0$$

  Because $y_1 \in C$ (no channel errors, $e = 0$).

- **Non-zero syndrome:** Single-bit error $e = \mathbf{1}\ 0\ 0\ 0\ 0\ 0\ 0\ 0\ 0\ 0$ into y_1:

$$y_2 = y_1 \oplus e \bmod 2 = \mathbf{0}\ 0\ 0\ 1\ 0 \mid 0\ 1\ 1\ 1\ 1$$

$$s_2 = y_2 \cdot H^T = e \cdot H^T \bmod 2 = 1\ 1\ 1\ 0\ 0$$

  Because $y_2 \notin C$, its syndrome equals the 1st row of H^T $(1\ 1\ 1\ 0\ 0)$.

3. Construct a different matrix $H^T \in \mathbb{F}_2^{10\times 5}$ such that a vector has zero syndrome if and only if it is a codeword. Compute the syndromes of the two vectors from Point 2.

- The equivalent Parity-Check Matrix H_1^T is obtained with an elementary column operation over $\mathbb{F}_2$:

$$\begin{aligned}\text{New Col}_1 = \text{Col}_1 \oplus \text{Col}_2 &= [1\ 0\ 0\ 1\ 1\ 1\ 0\ 0\ 0\ 0]^T + [1\ 1\ 0\ 0\ 1\ 0\ 1\ 0\ 0\ 0]^T \\ &= [\mathbf{0\ 1\ 0\ 1\ 0\ 1\ 1\ 0\ 0\ 0}]^T\end{aligned}$$

- Adding two columns is an elementary operation that keeps the columns linearly independent and preserves the rank of H^T, so the vectors giving a zero syndrome remain the exact same codewords.

$$H_1^T = \begin{bmatrix} \mathbf{0} & 1 & 1 & 0 & 0 \\ \mathbf{1} & 1 & 1 & 1 & 0 \\ \mathbf{0} & 0 & 1 & 1 & 1 \\ \mathbf{1} & 0 & 0 & 1 & 1 \\ \mathbf{0} & 1 & 0 & 0 & 1 \\ \mathbf{1} & 0 & 0 & 0 & 0 \\ \mathbf{1} & 1 & 0 & 0 & 0 \\ \mathbf{0} & 0 & 1 & 0 & 0 \\ \mathbf{0} & 0 & 0 & 1 & 0 \\ \mathbf{0} & 0 & 0 & 0 & 1 \end{bmatrix}$$

3. Construct a different matrix $H^T \in \mathbb{F}_2^{10\times 5}$ such that a vector has zero syndrome if and only if it is a codeword. Compute the syndromes of the two vectors from Point 2.

- **Zero syndrome:** Encode message $m = 1\ 0\ 0\ 1\ 0$ directly via $y_1 = m \cdot G_1 \bmod 2$:

$$y_1 = 1\ 0\ 0\ 1\ 0 \,|\, 0\ 1\ 1\ 1\ 1$$

$$s_1' = y_1 \cdot H_1^T \bmod 2 = 0\ 0\ 0\ 0\ 0$$

  Because $y_1 \in C$ (no channel errors, $e = 0$).

- **Non-zero syndrome:** Single-bit error $e = \mathbf{1}\ 0\ 0\ 0\ 0\ 0\ 0\ 0\ 0\ 0$ into y_1:

$$y_2 = y_1 \oplus e \bmod 2 = \mathbf{0}\ 0\ 0\ 1\ 0 \,|\, 0\ 1\ 1\ 1\ 1$$

$$s_2' = y_2 \cdot H_1^T = e \cdot H_1^T \bmod 2 = 0\ 1\ 1\ 0\ 0$$

  Because $y_2 \notin C$, its syndrome equals the 1st row of H_1^T $(0\ 1\ 1\ 0\ 0)$.

4. Construct a matrix $H^T \in \mathbb{F}_2^{10\times 7}$ such that a vector has zero syndrome if and only if it is a codeword. Compute the syndromes of the two vectors from Point 2.

- The equivalent Parity-Check Matrix H_2^T is obtained by appending two linearly dependent columns to H^T by duplicating Col_1 and Col_2:

$$H_2^T = \left[H^T \mid \text{Col}_1(H^T)\ \text{Col}_2(H^T)\right]$$

- Since the two added columns are copies of the first two, they are linearly dependent on them: the rank remains 5 (only redundant parity checks are added), so the vectors giving a zero syndrome remain the exact same codewords.

$$H_2^T = \begin{bmatrix} 1 & 1 & 1 & 0 & 0 & \mathbf{1} & \mathbf{1} \\ 0 & 1 & 1 & 1 & 0 & \mathbf{0} & \mathbf{1} \\ 0 & 0 & 1 & 1 & 1 & \mathbf{0} & \mathbf{0} \\ 1 & 0 & 0 & 1 & 1 & \mathbf{1} & \mathbf{0} \\ 1 & 1 & 0 & 0 & 1 & \mathbf{1} & \mathbf{1} \\ 1 & 0 & 0 & 0 & 0 & \mathbf{1} & \mathbf{0} \\ 0 & 1 & 0 & 0 & 0 & \mathbf{0} & \mathbf{1} \\ 0 & 0 & 1 & 0 & 0 & \mathbf{0} & \mathbf{0} \\ 0 & 0 & 0 & 1 & 0 & \mathbf{0} & \mathbf{0} \\ 0 & 0 & 0 & 0 & 1 & \mathbf{0} & \mathbf{0} \end{bmatrix}$$

4. Construct a matrix $H^T \in \mathbb{F}_2^{10\times 7}$ such that a vector has zero syndrome if and only if it is a codeword. Compute the syndromes of the two vectors from Point 2.

- **Zero syndrome:**

$$y_1 = 1\ 0\ 0\ 1\ 0\ |\ 0\ 1\ 1\ 1\ 1$$

$$s_1'' = y_1 \cdot H_2^T \bmod 2 = 0\ 0\ 0\ 0\ 0\ 0\ 0$$

  Because $y_1 \in C$ (no channel errors, $e = 0$).

- **Non-zero syndrome:** Single-bit error $e = \mathbf{1}\ 0\ 0\ 0\ 0\ 0\ 0\ 0\ 0\ 0$ into y_1:

$$y_2 = y_1 \oplus e \bmod 2 = \mathbf{0}\ 0\ 0\ 1\ 0\ |\ 0\ 1\ 1\ 1\ 1$$

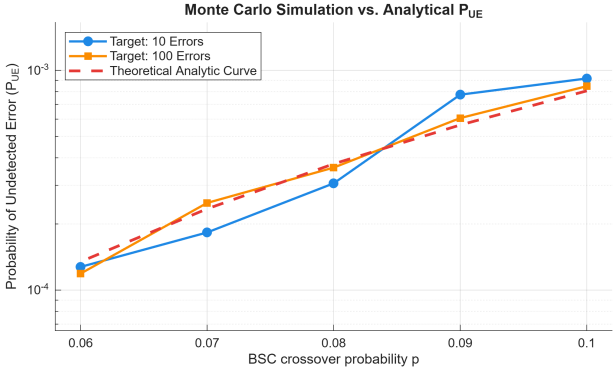
$$s_2'' = y_2 \cdot H_2^T = e \cdot H_2^T \bmod 2 = 1\ 1\ 1\ 0\ 0\ 1\ 1$$

  Because $y_2 \notin C$, its syndrome equals the 1st row of H_2^T $(1\ 1\ 1\ 0\ 0\ 1\ 1)$.

Exercise 1 - Part 4

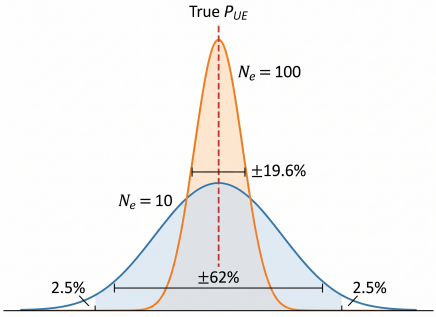
Monte Carlo Simulation

1. Plot the two Monte Carlo curves (10 and 100 undetected errors) together with the analytical curve



2. Explain what a confidence interval is and how it could be used to assess the reliability of the Monte Carlo estimates

- A confidence interval is an estimated range of values $[\hat{P}_{UE} - \Delta, \hat{P}_{UE} + \Delta]$, computed from a finite simulation sample, used to quantify the statistical uncertainty of a point estimate $\hat{P}_{UE}$.
- At a 95% confidence level, repeating the simulation many times would produce intervals containing the true P_{UE} in 95% of runs.
- For rare events ($P_{UE} \ll 1$), the relative CI half-width depends strictly on the target error count N_e: $\Delta/P_{UE} \approx 1.96/\sqrt{N_e}$.
- **Blue Curve ($N_e = 10$):** Relative uncertainty is $\pm 62\%$, explaining why the blue markers swing noticeably above ($p = 0.09$) and below ($p = 0.07$) the red dashed curve.
- **Orange Curve ($N_e = 100$):** Relative uncertainty drops to $\pm 19.6\%$ ($3.16\times$ tighter), keeping every orange point tightly aligned with the theoretical curve.



3. Compare the results, discussing statistical fluctuations, agreement with the analytical curve, and the trade-off between simulation accuracy and computational time.

- **Target 10 (Blue):** Shows significant random variance across $p \in [0.06, 0.10]$.
- **Target 100 (Orange):** Closely matches the Theoretical Analytic Curve across the entire range.
- Improving accuracy from 10 to 100 undetected errors requires $10\times$ more transmitted codewords.
- At $p = 0.06$ ($P_{UE} \approx 1.3 \times 10^{-4}$), collecting 100 undetected errors requires simulating roughly 7.7×10^5 codewords—showing how runtime grows inversely with P_{UE} as p decreases.

3. Explain the advantages of having an analytical expression for P_{UE}.

- **Analytical Expression** ($n = 10$, $A_4 = 15$, $A_6 = 15$, $A_{10} = 1$):

$$P_{UE} = \sum_{w=1}^{n} A_w \, p^w (1-p)^{n-w} = 15\, p^4 (1-p)^6 + 15\, p^6 (1-p)^4 + p^{10}$$

- The advantages are:
  - Eliminates random Monte Carlo sampling noise, because it is a deterministic formula;
  - Evaluates P_{UE} in $n = 10$ operations for any p, whereas Monte Carlo runtime explodes as $1/P_{UE}$ when $p \to 0$;
  - Relates error-detection performance directly to the weight enumerator A_w (multiplicity), enabling instant comparison across different linear codes.

Exercise 1 - Part 5

Retransmissions (ARQ)

1. Write the expression of $p_0 = P_{\mathbf{COR},i=1}$, $p_1 = P_{\mathbf{UE},i=1}$, $p_2 = P_{\mathbf{REJ},i=1}$.

- **Correct Reception Probability** ($n = 10$):

$$p_0 = P_{\mathrm{COR},i=1} = (1-p)^n = (1-p)^{10}$$

- **Undetected Error Probability** ($A_4 = 15$, $A_6 = 15$, $A_{10} = 1$):

$$p_1 = P_{\mathrm{UE},i=1} = \sum_{w=1}^{n} A_w \, p^w (1-p)^{n-w} = 15\,p^4(1-p)^6 + 15\,p^6(1-p)^4 + p^{10}$$

- **Block Rejection Probability**:

$$\begin{aligned} p_2 = P_{\mathrm{REJ},i=1} &= 1 - p_0 - p_1 \\ &= 1 - (1-p)^{10} - \left[15\,p^4(1-p)^6 + 15\,p^6(1-p)^4 + p^{10}\right] \end{aligned}$$

2. Write the analytical expression of $P_{\text{COR},i}$ for $i = 1, 2, 3$.

- **1st Attempt ($i = 1$):**

$$P_{\text{COR},1} = p_0 = (1-p)^n = (1-p)^{10}$$

- **2nd Attempt ($i = 2$):** (1 rejection followed by 1 correct reception)

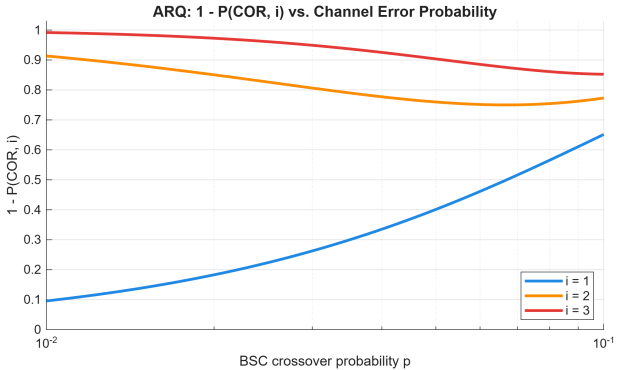
$$P_{\text{COR},2} = p_2 \cdot p_0 = (1 - p_0 - p_1)p_0 = \left[1 - (1-p)^{10} - p_1\right](1-p)^{10}$$

- **3rd Attempt ($i = 3$):** (2 consecutive rejections followed by 1 correct reception)

$$P_{\text{COR},3} = p_2^2 \cdot p_0 = (1 - p_0 - p_1)^2 p_0 = \left[1 - (1-p)^{10} - p_1\right]^2 (1-p)^{10}$$

where $p_1 = 15\,p^4(1-p)^6 + 15\,p^6(1-p)^4 + p^{10}$.

3. Plot $1 - P(\mathbf{COR}, i)$ versus p over $10^{-2} \leq p \leq 10^{-1}$ for $i = 1, 2, 3$ on the same figure



Exercise 1 - Part 6

Discussion on ARQ

1. Explain if allowing more transmission attempts increases the probability of correct reception

- **Yes:** Allowing more attempts i increases the total probability of correct reception $P(\text{COR}, i)$:

$$P(\text{COR}, i) = \sum_{k=1}^{i} p_2^{k-1} p_0 = p_0 \frac{1 - p_2^i}{1 - p_2}$$

- Whenever a block is rejected (with probability $p_2 < 1$), the transmitter gets another chance to send it correctly, adding a positive term $p_2^i p_0 > 0$.
- As i grows, $P(\text{COR}, i)$ approaches its maximum limit $\frac{p_0}{p_0 + p_1}$ (meaning the failure curve $1 - P(\text{COR}, i)$ shifts downward, as seen in the previous plot).

2. Explain if retransmissions can eliminate undetected errors.

- **No:** Retransmissions cannot eliminate undetected errors.
- ARQ only requests a retransmission when an error is detected ($s \neq 0$, probability p_2). If channel noise turns the sent codeword into another valid codeword ($s = 0$, probability p_1), the receiver accepts it and never asks for a retransmission.
- Allowing more attempts i actually slightly increases the overall undetected error probability:

$$P(\text{UE}, i) = p_1 \sum_{k=0}^{i-1} p_2^k = p_1 \frac{1 - p_2^i}{1 - p_2} \xrightarrow{i \to \infty} \frac{p_1}{p_0 + p_1}$$

3. Discuss the trade-off between reliability and delay.

- Allowing more transmission attempts i improves reliability: $P(\text{COR}, i)$ increases and the probability of dropping a packet after i failed attempts (p_2^i) decreases.
- However, each retransmission adds extra delay (waiting for feedback and re-sending the block) and lowers the effective throughput because the channel is used multiple times for the same message.
- Small i: low latency and fast delivery, but higher packet loss when the channel is noisy.
- Large i: high reliability (up to the undetected error floor), at the cost of higher average and worst-case delay, especially at high p.

4. Compute the expected number of transmissions for a maximum of $i = 1, 2, 3$ attempts, and generalize the result to an arbitrary i. (Hint: determine the probabilities of performing exactly 1, 2, or 3 transmissions, as applicable, and use them to compute the expected value.) What happens as $i \to \infty$? Discuss the result.

- A block is accepted with probability $1 - p_2 = p_0 + p_1$ and rejected (triggering a retransmission) with probability p_2:
  - $i = 1$: $P(N = 1) = 1 \implies E[N]_1 = 1$
  - $i = 2$: $P(N = 1) = 1 - p_2,\ P(N = 2) = p_2 \implies E[N]_2 = 1(1 - p_2) + 2p_2 = 1 + p_2$
  - $i = 3$:
    $P(N = 1) = 1 - p_2,\ P(N = 2) = p_2(1 - p_2),\ P(N = 3) = p_2^2 \implies E[N]_3 = 1 + p_2 + p_2^2$
- Generalizing for an arbitrary maximum number of attempts i:

$$E[N]_i = \sum_{k=0}^{i-1} p_2^k = \frac{1 - p_2^i}{1 - p_2}$$

- As $i \to \infty$, since $0 \leq p_2 < 1$, $p_2^i \to 0$ and the expected number of transmissions converges to a finite value:

$$\lim_{i \to \infty} E[N]_i = \frac{1}{1 - p_2} = \frac{1}{p_0 + p_1}$$

Even with unlimited attempts, the average delay remains finite (close to 1 for small p) because rejections become geometrically unlikely.

5. Use your result from Question 4 to support your discussion in Question 3.

- The expected number of transmissions $E[N]_i = \frac{1-p_2^i}{1-p_2}$ directly quantifies the average transmission delay and channel usage, while $P(\text{COR}, i) = p_0 \frac{1-p_2^i}{1-p_2}$ quantifies reliability.
- Both quantities scale with the same factor $\frac{1-p_2^i}{1-p_2}$: increasing i reduces packet loss (p_2^i) and raises reliability, at the price of a higher average number of transmissions $E[N]_i$ (lower throughput and higher latency).
- At low p ($p_2 \ll 1$), higher-order terms p_2^k vanish rapidly, so $E[N]_i \approx 1$ even for large i: we gain reliability with almost zero extra average delay.
- At high p (p_2 closer to 1), increasing i significantly boosts $E[N]_i$ toward $\frac{1}{1-p_2}$, making the latency penalty substantial.

Exercise 2a

CRC Codes and Weight Distribution

1. Choose a primitive binary polynomial $p(D)$ of degree r.

- For this part, we selected the primitive binary polynomial of degree $r_p = 3$:

$$p(D) = D^3 + D + 1$$

2. Construct the generator polynomial $g(D) = (D+1)p(D)$.

- The generator polynomial $g(D)$ is obtained by multiplying the primitive polynomial $p(D)$ by $(D+1)$:

$$g(D) = (D+1)p(D) = (D+1)(D^3 + D + 1) = D^4 + D^3 + D^2 + 1$$

5. Construct a table containing the four information vectors v, the corresponding information polynomials $v(D)$, the codeword polynomials $c(D)$, the codeword vectors c, and their Hamming weights.

Information vectors v	Information polynomials $v(D)$	Codeword polynomials $c(D)$	Codeword vectors c	Hamming weight w_H
0 0	0	0	0 0 0 0 0 0	0
0 1	1	$D^4 + D^3 + D^2 + 1$	0 1 1 1 0 1	4
1 0	D	$D^5 + D^2 + D + 1$	1 0 0 1 1 1	4
1 1	$D + 1$	$D^5 + D^4 + D^3 + D$	1 1 1 0 1 0	4

6. Compute the weight distribution coefficients A_i

Weight coefficients A_i
$A_0 = 1$
$A_1 = 0$
$A_2 = 0$
$A_3 = 0$
$A_4 = 3$
$A_5 = 0$
$A_6 = 0$

- The weight distribution shows that the code contains one codeword of weight 0 and three codewords of weight 4.
- All other weight values have a zero coefficient.

7. Comment on the results

- The code has a minimum Hamming distance of $d_{\text{min}} = 4$, which guarantees the detection of up to 3 bit errors and the correction of a single-bit error.
- Another important property is that all codewords have an even Hamming weight ($w_H \in \{0, 4, 6\}$). This is ensured by the presence of the factor $(D + 1)$ in the generator polynomial, which forces every codeword to contain an even number of ones.
- As a result, the code can detect any channel error involving an odd number of bit flips, since such errors change the parity of the received word, making it immediately recognizable as an invalid codeword.

9. Present a table reporting the weight distribution coefficients A_i for each value of k considered

$\mathbf{k = 2 \quad (n = 6)}$	$\mathbf{k = 3 \quad (n = 7)}$	$\mathbf{k = 4 \quad (n = 8)}$
$A_0 = 1$	$A_0 = 1$	$A_0 = 1$
$A_1 = 0$	$A_1 = 0$	$A_1 = 0$
$A_2 = 0$	$A_2 = 0$	$A_2 = 1$
$A_3 = 0$	$A_3 = 0$	$A_3 = 0$
$A_4 = 3$	$A_4 = 7$	$A_4 = 11$
$A_5 = 0$	$A_5 = 0$	$A_5 = 0$
$A_6 = 0$	$A_6 = 0$	$A_6 = 3$
—	$A_7 = 0$	$A_7 = 0$
—	—	$A_8 = 0$

10. Explain mathematically why weight-2 codewords appear for exactly that value of k.

- A codeword with Hamming weight 2 can be expressed as: $c(D) = D^i + D^j = D^j(D^L + 1)$ where $L = i - j$ is the distance between the two non-zero bits.
- Since a valid codeword must be divisible by the generator polynomial $g(D) = (D + 1)p(D)$, with $p(D) = D^3 + D + 1$, the polynomial $p(D)$ must divide $D^L + 1$.
- Being a primitive polynomial of degree 3, $p(D)$ has order: $N = 2^3 - 1 = 7$.
- Therefore, the smallest possible distance is $L = 7$. The corresponding minimum codeword is: $c(D) = D^7 + 1$.
- Its degree is 7, requiring a codeword length of at least $n = 8$. Since $n = k + r$, with $r = 4$: $k + 4 \geq 8 \implies k \geq 4$.
- **Conclusion:** Weight-2 codewords first appear at $k = 4$, because this is the minimum information length that allows two non-zero bits separated by 7 positions.

Exercise 2b

CRC Codes – Burst Error Detection

1. What is the maximum burst length L that is certainly detected by this CRC code?

- A CRC code with r redundancy bits guarantees the detection of all burst errors of length:

$$L \leq r$$

- Since the generator polynomial has degree $r = 4$, the code is guaranteed to detect all burst errors of length up to:

$$L_{\text{max}} = 4 \text{ bits}$$

- Therefore, this CRC code detects with certainty any burst error of length 4 bits or less.

2. Provide a mathematical proof of your answers using the polynomial representation of burst errors

- A burst error of length L can be represented as $e(D) = D^j b(D)$, where $b(D)$ is the burst polynomial with degree $\deg b(D) = L - 1$, and D^j represents the position of the burst within the transmitted block.
- An error goes undetected if and only if it is divisible by the generator polynomial $g(D)$. Since $g(0) = 1$, D^j and $g(D)$ are relatively prime. Therefore:

$$g(D) \mid e(D) \iff g(D) \mid b(D)$$

- For burst lengths $L \leq r$, we have:

$$\deg b(D) = L - 1 \leq r - 1 < r = \deg g(D)$$

- Thus, $b(D)$ cannot be a non-zero multiple of $g(D)$. Every burst error of length $L \leq r = 4$ is guaranteed to be detected by the CRC code.

3. Find two burst error patterns e_1 and e_2

- **1. Detected burst error ($L = 4$):**

$$e_1 = [1\ 0\ 1\ 1\ 0\ 0\ 0\ 0]^T$$

Its polynomial representation is:

$$e_1(D) = D^3 + D + 1$$

- **2. Undetected burst error ($L = 5$):**

$$e_2 = [1\ 1\ 1\ 0\ 1\ 0\ 0\ 0]^T$$

Its polynomial representation is:

$$e_2(D) = D^4 + D^3 + D^2 + 1 = g(D)$$

4. Explain mathematically why one error is detected and the other is not

- The first error polynomial is $e_1(D) = D^3 + D + 1$. Since its degree is smaller than the degree of the generator polynomial ($\deg e_1 = 3 < \deg g = 4$), it cannot be divisible by $g(D)$.
- Therefore, the division produces a non-zero remainder:

$$\text{rem}\left(\frac{e_1(D)}{g(D)}\right) = D^3 + D + 1 \neq 0$$

  The syndrome is non-zero, so the CRC detects the error.
- The second error polynomial is $e_2(D) = D^4 + D^3 + D^2 + 1 = g(D)$. Since the error polynomial coincides with the generator polynomial, the division gives:

$$\text{rem}\left(\frac{e_2(D)}{g(D)}\right) = 0$$

  The syndrome is zero, meaning that the error is indistinguishable from a valid codeword.

5. Choose a codeword and verify your results

- The selected information sequence is:

$$v = [1\ 0\ 1\ 1]^T$$

- Using the CRC encoder with generator polynomial $g(D) = D^4 + D^3 + D^2 + 1$, the resulting codeword is:

$$c = [1\ 0\ 1\ 1\ 0\ 0\ 1\ 0]^T$$

- Two transmission scenarios were tested:
  - **Burst error (e_1), $L = 4$:** detected by the CRC (non-zero syndrome).
  - **Burst error (e_2), $L = 5$:** not detected by the CRC (zero syndrome).

Exercise 3a

Decision Regions property for a linear code

1. Prove that, for every codeword $c \in C$, $D(c) = c + D(0)$ (1/2).

- Consider a binary linear code $C(n, k)$ with minimum-distance decoding. Writing $y = c + e$ (so $y + c = e$) and $c' = c + c''$ (so $c' + c = c'' \in C$ since C is linear), we develop $D(c)$ in four steps:

$$\begin{aligned} D(c) &= \{y \in \mathbb{F}_2^n : d_H(y, c) \leq d_H(y, c'),\ \forall c' \in C\} \\ &= \{c + e \in \mathbb{F}_2^n : d_H(c + e + c, c + c) \leq d_H(c + e + c, c' + c),\ \forall c' \in C\} \\ &= \{c + e \in \mathbb{F}_2^n : d_H(e, 0) \leq d_H(e, c''),\ \forall c'' \in C\} \\ &= \{c + e \in \mathbb{F}_2^n : w_H(e) \leq d_H(e, c''),\ \forall c'' \in C\} \end{aligned}$$

1. Prove that, for every codeword $c \in C$, $D(c) = c + D(0)$ (2/2).

- Similarly, developing $D(0)$ with $y = c + e$ and $d_H(y, 0) = w_H(y)$:

$$\begin{aligned} D(0) &= \{y \in \mathbb{F}_2^n : d_H(y, 0) \leq d_H(y, c'),\ \forall c' \in C\} \\ &= \{c + e \in \mathbb{F}_2^n : w_H(c + e) \leq d_H(c + e, c'),\ \forall c' \in C\} \\ &\text{with } c = 0, \text{ so} \\ &= \{0 + e \in \mathbb{F}_2^n : w_H(0 + e) \leq d_H(0 + e, c'),\ \forall c' \in C\} \\ &= \{e \in \mathbb{F}_2^n : w_H(e) \leq d_H(e, c'),\ \forall c' \in C\} \end{aligned}$$

- Writing $D(0) = \{e_0, e_1, \ldots, e_{N-1}\}$, since every element in $D(c)$ has the form $y = c + e$:

$$D(c) = \{c + e_0, c + e_1, \ldots, c + e_{N-1}\} = \{e_0, e_1, \ldots, e_{N-1}\} + c = D(0) + c = c + D(0)$$

2. Discuss the meaning of this property

- All decision regions $D(c)$ have the exact same shape and size, since each one is simply $D(0)$ shifted by the codeword c.
- Because every codeword has the exact same set of correctable error patterns around it, the word error probability $P_W(E)$ over a BSC does not depend on the transmitted codeword (so in analysis or simulations we can just assume the all-zero codeword 0 was sent).
- Instead of storing 2^k separate decision regions, the decoder only needs to store $D(0)$ (the coset leaders) and subtract the error vector $e \in D(0)$ from the received word y.

Exercise 3b

Decision Regions of a linear code

1. Choose a non-systematic generator matrix G of rank 3.

- **Non-Systematic Generator Matrix G:**

$$G = \begin{bmatrix} 1 & 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 0 \end{bmatrix}$$

- Dimensions: 3×6 ($k = 3$ information bits, $n = 6$ coded bits)
- Does not contain an identity submatrix I_3 so is in Non-systematic form.

5. Write all the decision regions $D(c)$

- $D(c_1)$: Codeword **000000**:
  $\{\mathbf{000000}, 000001, 000010, 000100, 001000, 100000, 000110, 001100, 100010, 101000\}$
- $D(c_2)$: Codeword **111110**:
  $\{\mathbf{111110}, 111111, 111100, 111010, 110110, 011110, 111000, 110010, 011100, 010110\}$
- $D(c_3)$: Codeword **011011**:
  $\{\mathbf{011011}, 011010, 011001, 011111, 010011, 111011, 011101, 010111, 111001, 110011\}$
- $D(c_4)$: Codeword **100101**:
  $\{\mathbf{100101}, 100100, 100111, 100001, 101101, 000101, 100011, 101001, 000111, 001101\}$
- $D(c_5)$: Codeword **110101**:
  $\{\mathbf{110101}, 110100, 110111, 110001, 111101, 010101, 110011, 111001, 010111, 011101\}$
- $D(c_6)$: Codeword **001011**:
  $\{\mathbf{001011}, 001010, 001001, 001111, 000011, 101011, 001101, 000111, 101001, 100011\}$
- $D(c_7)$: Codeword **101110**:
  $\{\mathbf{101110}, 101111, 101100, 101010, 100110, 001110, 101000, 100010, 001100, 000110\}$
- $D(c_8)$: Codeword **010000**:
  $\{\mathbf{010000}, 010001, 010010, 010100, 011000, 110000, 010110, 011100, 110010, 111000\}$

5. Write all the decision regions $D(c)$

- Here, each region includes all vectors closest to c_i without arbitrary tie-breaking:
  - 1 vector at $d_H = 0$ (the codeword itself)
  - 6 vectors at $d_H = 1$ (all single-bit errors, uniquely decoded since $d_{\text{min}} = 3$)
  - 3 vectors at $d_H = 2$ (boundary vectors equidistant from another codeword)
- Overlapping regions ($8 \times 10 = 80 > 64$), where 24 boundary vectors appear in multiple regions (3 per codeword).

6. Verify that $D(c) = c + D(0), \quad \forall c \in C$

- Verification Logic implemented in MATLAB:
  - Extract the reference region D_0 corresponding to $c_1 = 0\ 0\ 0\ 0\ 0\ 0$
  - Compute the shifted set $D_C = c_i \oplus D_0 \bmod 2$
  - Compare sorted sets with the command: `isequal(sortrows(y_region), sortrows(D_C))`
    $\rightarrow$ Rule Verified for all $i \in \{1, \ldots, 8\}$

Exercise 3c

Decision Regions of Nonlinear Codes

1. Choose a set of four distinct codewords of length $n = 4$ bits, including 0, that does not form a linear code

- A set of 4 distinct binary codewords of length $n = 4$ containing the all-zero vector 0:

$$C = \begin{bmatrix} 0000 \\ 0010 \\ 0011 \\ 1100 \end{bmatrix} \begin{matrix} \leftarrow c_1 = 0 \\ \leftarrow c_2 \\ \leftarrow c_3 \\ \leftarrow c_4 \end{matrix}$$

- $n = 4$ bits, $|C| = 4$ ($k = 2$ information bits), contains 0 ($c_1 = 0\ 0\ 0\ 0$).
- A binary code C is linear if and only if it is closed under modulo-2 addition ($\oplus$): $\forall c_i, c_j \in C \implies c_i \oplus c_j \in C$.
- Counterexample ($c_2 \oplus c_4$): $c_2 \oplus c_4 = 0\ 0\ 1\ 0 \oplus 1\ 1\ 0\ 0 = 1\ 1\ 1\ 0 \notin C$.
- Since 1 1 1 0 is not in C, closure fails $\implies$ C is a non-linear code.

3. Write $D(c)$ for each of the four codewords

- $D(c_1)$: Codeword **0000** (6 vectors): $\{\mathbf{0000}, 0001, 0100, 0101, 1000, 1001\}$
- $D(c_2)$: Codeword **0010** (3 vectors): $\{\mathbf{0010}, 0110, 1010\}$
- $D(c_3)$: Codeword **0011** (7 vectors): $\{\mathbf{0011}, 0001, 0101, 0111, 1001, 1011, 1111\}$
- $D(c_4)$: Codeword **1100** (8 vectors): $\{\mathbf{1100}, 0100, 0101, 1000, 1001, 1101, 1110, 1111\}$

4. Check whether the following property holds: $D(c) = c + D(0), \quad \forall c \in C$

- **Verification of $D(c) = c + D(0)$ for the Non-Linear Code**
- Counterexample ($c_2 = 0\ 0\ 1\ 0$):
  - $c_2 \oplus D_0$ (6 vectors): $\{0010, 0011, 0110, 0111, 1010, 1011\}$
  - Actual Decision Region $D(c_2)$ (3 vectors): $\{0010, 0110, 1010\}$
- Vectors ending in 1 (e.g., 0011) belong to C itself ($c_3 = 0\ 0\ 1\ 1$), so they are captured by $D(c_3)$ instead of $D(c_2)$.

5. Comment the results

- Without algebraic closure ($c_i \oplus c_j \in C$), neighboring codewords sit at irregular distances, breaking geometric symmetry and making $D(c) = c + D(0)$ invalid.
- The property holds exclusively for the first codeword $c_1 = 0\ 0\ 0\ 0$ as a trivial identity:

$$D(c_1) = c_1 + D(0) \implies D(0) = 0 \oplus D(0)$$

- Adding 0 leaves every vector unchanged, so this case does not rely on the linearity (closure) of the code. For every non-zero codeword (c_2, c_3, c_4), the property fails.

Exercise 4

Error Correction with the Hamming (7,4) Code

Exercise 4a

Analytical Expression of $P_W(E)$

1. Verify that the code is perfect.

- For the Hamming $(7,4)$ code ($n = 7$, $k = 4$, $d_{\min} = 3$), the error-correction capability is:

$$t = \left\lfloor \frac{d_{\min} - 1}{2} \right\rfloor = \frac{3-1}{2} = 1 \text{ bit}$$

- A code is **perfect** if the disjoint Hamming spheres $S(c,t)$ around the 2^k codewords completely cover $\mathbb{F}_2^n$, i.e., the Hamming bound holds with equality:

$$2^k \sum_{i=0}^{t} \binom{n}{i} = 2^n$$

- Since $|S(c,1)| = \binom{7}{0} + \binom{7}{1} = 1 + 7 = 8$ and there are $2^4 = 16$ codewords:

$$16 \times 8 = 128 = 2^7$$

- Equality holds, so **the Hamming $(7,4)$ code is perfect**.

2. Derive the analytical expression for the word error probability $P_W(E)$ as a function of p.

- Over a BSC with crossover probability p, minimum Hamming distance decoding maps $y = c \oplus e$ to the closest codeword.
- Since the $(7, 4)$ code is **perfect** ($t = 1$), the decision regions are $D(c) = S(c, 1)$: the decoder corrects **all and only** error patterns with $w_H(e) \leq 1$.
- The probability of correct word decoding is therefore:

$$P_W(C) = \sum_{i=0}^{t} \binom{n}{i} p^i (1-p)^{n-i} = (1-p)^7 + 7p(1-p)^6$$

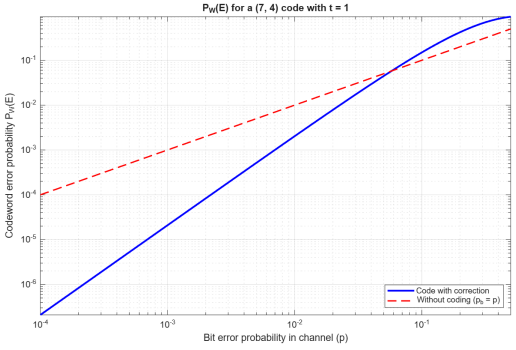
- Hence, the exact word error probability $P_W(E) = 1 - P_W(C)$ is:

$$P_W(E) = 1 - (1-p)^7 - 7p(1-p)^6 = \sum_{w=2}^{7} \binom{7}{w} p^w (1-p)^{7-w}$$

- For small p ($p \ll 1$), the asymptotic behavior is $P_W(E) \approx \binom{7}{2} p^2 = 21p^2$.

3. Plot the theoretical $P_W(E)$ versus p.

- Theoretical word error probability $P_W(E) = 1 - (1-p)^7 - 7p(1-p)^6$ of the Hamming $(7, 4)$ code ($t = 1$) over $p \in [10^{-4}, 0.5]$, compared with uncoded single-bit transmission ($p_b = p$).



Exercise 4b

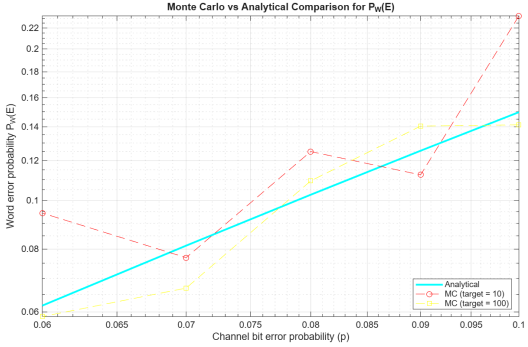
Simulation

2. Explain why you focused on this region of the curve.

- We focused on the interval $p \in [0.06, 0.10]$ because for smaller values of p the word error probability $P_W(E)$ drops very quickly ($\propto p^2$).
- Since the number of simulated codewords needed to observe N_e errors scales as $N_e/P_W(E)$, collecting 100 word errors at $p = 0.06$ ($P_W(E) \approx 0.064$) takes around 1.56×10^3 transmissions, whereas at $p = 10^{-4}$ ($P_W(E) \approx 2.1 \times 10^{-7}$) it would require almost 5×10^8 transmissions, making the simulation too slow.
- Moreover, around $p \approx 0.056$ the $P_W(E)$ curve crosses the uncoded reference $p_b = p$, so this region shows both the transition where coding becomes advantageous and the theoretical $\propto p^2$ slope on a log-log plot.

3. Plot the analytical and simulated curves on the same graph

- Comparison between the theoretical $P_W(E)$ curve and the Monte Carlo simulations with a stopping target of 10 and 100 word errors over $p \in [0.06, 0.10]$.



4. Comment on the results

- With a target of **10 word errors**, the simulated curve shows strong fluctuations around the analytical curve and is not even monotonic (e.g., jumping to ≈ 0.23 at $p = 0.10$ and rising again at $p = 0.06$) because the sample size is too small to average out random channel noise.
- Increasing the target to **100 word errors** reduces the statistical variance by a factor of 10, making the empirical curve much smoother and closely aligned with the theoretical curve across the whole interval.
- As p decreases, $P_W(E)$ decreases following the theoretical $\propto p^2$ slope on the log-log plot, confirming that double-bit errors dominate in this regime.
- Overall, there is a direct trade-off between accuracy and simulation runtime: increasing the target from 10 to 100 errors yields a much more reliable estimate at the cost of $10\times$ more simulated transmissions.

Exercise 4c

Another (8,4) Code

1. Determine the error-correction capability t of the new code

- From the systematic generator matrix $G = [I_4 \mid A] \in \mathbb{F}_2^{4\times 8}$, the code has parameters $n = 8$ and $k = 4$.
- The minimum Hamming weight among all non-zero codewords is $d_{\min} = 3$ (e.g., the third row of G, 0010 0011, has weight 3).
- The error-correction capability is therefore:

$$t = \left\lfloor \frac{d_{\min} - 1}{2} \right\rfloor = \frac{3-1}{2} = 1 \text{ bit}$$

2. Verify if it is perfect

- For $n = 8$ and $t = 1$, a Hamming sphere centered at each codeword contains:

$$|S(c,1)| = \binom{8}{0} + \binom{8}{1} = 1 + 8 = 9 \text{ vectors}$$

- Multiplying by the $2^k = 2^4 = 16$ codewords gives:

$$16 \times 9 = 144 \neq 2^8 = 256$$

- Since the Hamming bound does not hold with equality (112 vectors in $\mathbb{F}_2^8$ fall outside the spheres of radius 1), **this** $(8,4)$ **code is not perfect**.

3. Derive analytically $P_W(E)$.

- With bounded-distance decoding ($t = 1$), the decoder corrects error patterns of weight 0 or 1, so the probability of correct decoding is:

$$P_W(C) = (1-p)^8 + 8p(1-p)^7$$

- Therefore, the word error probability is:

$$P_W(E) = 1 - P_W(C) = 1 - (1-p)^8 - 8p(1-p)^7 = \sum_{w=2}^{8} \binom{8}{w} p^w (1-p)^{8-w}$$

- Since the code is not perfect, this formula is exact only for bounded-distance decoding (which corrects up to $t = 1$ error), while under complete minimum-distance decoding some weight-2 errors could also be corrected.

5. Compare the results obtained for the two codes and comment on any differences.

- Both codes have $k = 4$, $d_{\min} = 3$, and $t = 1$, but the $(8, 4)$ code uses an extra parity bit ($n = 8$ instead of $n = 7$), lowering the code rate from $R = 4/7 \approx 0.57$ to $R = 4/8 = 0.5$.
- Since n is larger while $t = 1$ stays the same, there are more double-bit error combinations ($\binom{8}{2} = 28$ vs. $\binom{7}{2} = 21$), so the $(8, 4)$ code has a slightly higher word error probability ($P_W(E) \approx 28p^2$ vs. $\approx 21p^2$ for small p).
- Unlike the Hamming $(7, 4)$ code, which is perfect ($16 \times 8 = 128 = 2^7$), the $(8, 4)$ code is not perfect ($16 \times 9 = 144 < 256$), leaving 112 vectors outside the Hamming spheres of radius 1.

Exercise 5.1

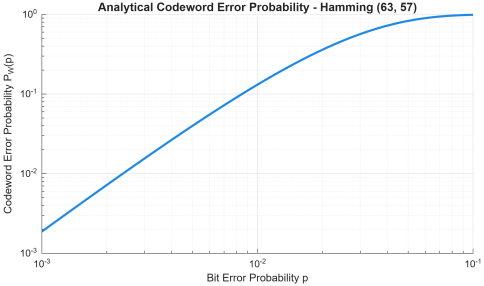
Syndrome Decoding

2. Derive and plot the analytical codeword error probability $P_W(p)$ for $p \in [10^{-3}, 10^{-1}]$.

- The $(63, 57)$ binary Hamming code has $d_{\min} = 3$, so it corrects up to $t = \lfloor (d_{\min} - 1)/2 \rfloor = 1$ bit error per codeword.
- A codeword error occurs whenever 2 or more bit errors happen over $n = 63$ bits:

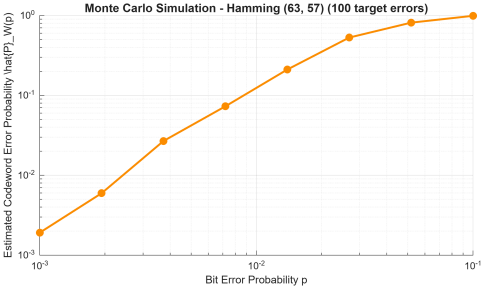
$$P_W(p) = 1 - (1-p)^{63} - 63\,p(1-p)^{62}$$

- For small p ($p \approx 10^{-3}$),
  $P_W(p) \approx \binom{63}{2} p^2 = 1953\,p^2 \approx 1.95 \times 10^{-3}$.
- As p approaches 10^{-1}, multiple errors become dominant and $P_W(p)$ rises steeply toward 1 ($P_W(0.1) \approx 0.99$).



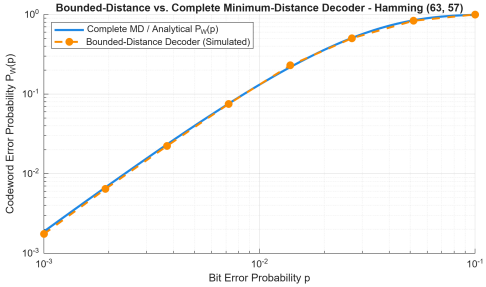
3. Simulate the bounded-distance decoder over the BSC for 8 values of $p \in [10^{-3}, 10^{-1}]$ (stopping at 100 codeword errors per point) and plot the estimated $\hat{P}_W(p)$.

- For each of the 8 logarithmically spaced values of p, random codewords $c = uG \bmod 2$ are transmitted over a BSC until 100 decoding errors occur.
- At the receiver, the syndrome $s = yH^T \bmod 2$ is computed:
  - if $s = 0$, no error is assumed;
  - if $s \neq 0$, the bit corresponding to the matching column of H is flipped.
- A codeword error is counted whenever $\hat{c} \neq c$, yielding the Monte Carlo estimate $\hat{P}_W(p) = 100/N_{\text{words}}$.
- Stopping at 100 errors ensures a consistent relative confidence interval across all simulated points, even at low p.



5. Compare the performance of the bounded-distance decoder with that of a complete minimum-distance decoder for this code.

- The $(63, 57)$ binary Hamming code is a perfect code: the 2^{57} disjoint Hamming spheres of radius $t = 1$, each containing $1 + 63 = 2^6$ vectors, completely partition $\{0, 1\}^{63}$ ($2^{57} \cdot 2^6 = 2^{63}$).
- Every possible 6-bit syndrome maps uniquely to an error pattern of weight $w \leq 1$, so no received vector ever falls outside the decoding spheres and decoding failures never occur.
- Thus, the bounded-distance decoder is identical to a complete minimum-distance decoder, as confirmed by the exact overlap between the simulated curve and the analytical bound.



7. Discuss how changing (n, k) affects error-correction capability and code rate, and whether any (n, k) pair can be chosen for a binary Hamming code.

- No: a binary Hamming code cannot have an arbitrary (n, k) pair. Its parameters are strictly constrained to $(n, k) = (2^r - 1,\ 2^r - 1 - r)$ for an integer $r \geq 3$.
- Every binary Hamming code has $d_{\min} = 3$, meaning its error-correction capability is always fixed at $t = 1$ bit error per codeword regardless of r (to achieve $t > 1$, other code families such as BCH codes are required).
- Increasing r raises the code rate $R = k/n = 1 - \frac{r}{2^r - 1}$ (less redundancy overhead), but since $t = 1$ remains fixed over a longer block n, multiple errors become more likely and $P_W(p) \approx \binom{n}{2} p^2$ degrades.

Exercise 6

Properties of m-sequences

Exercise 6 - Sequence

1. **Choose a random seed and write it.**

3. Plot the first 20 bits of the bipolar sequence ($0 \rightarrow -1$ and $1 \rightarrow +1$)

Exercise 6 - Sequence

3. **Comment the result.**

Exercise 6 - Number of zero/ones, number of transitions

1. Write a table with N_1 and N_0 (number of bits equal to 1 and 0), N_T and N_{NT} (number of transitions and no transitions, view the sequence as periodic).

Exercise 6 - Number of zero/ones, number of transitions

2. Check if the properties of an ideal binary sequences are verified and discuss the result

Exercise 6 - Runs

1. Write a table with the values of $N_R(T)$ (total number of runs), $N_R(i)$ (number of runs of length i), $N_{R0}(i)$ and $N_{R1}(i)$ (number of runs of zeros and ones of length i)

2. Plot the run lengths vs. their starting point in the sequence (use different colors for 0 and 1 runs)

Exercise 6 - Runs

3. Compare against the run properties of an ideal binary random sequence and discuss the results.

Exercise 6 - Autocorrelation

1. **Plot the autocorrelation**

Exercise 6 - Autocorrelation

2. Compare its properties with those of an ideal binary random sequences.

Exercise 6 - Truncated sequences - MPSL

5. Plot $MPSL(z)$ vs. z

6. Comment the results

Exercise 6 - Truncated sequences - seed

4. Plot it

5. Repeat and plot with the same z but another starting seed

Exercise 6 - Truncated sequences - seed

6. Comment the results

Grazie!